



RUNBOOK

VIRTUAL CONTROL

VMWARE CLOUD FOUNDATION SOLUTIONS

Implementing Capacity Reporting & Forecast Views in VCF Operations 9

End-to-end: tune the capacity policy, engineer the reporting super metrics, build forecast Trend views and Time-Remaining lists, surface them on a dashboard, and ship a scheduled PDF/CSV report — plus What-If planning. Every navigation path cited to [Broadcom TechDocs](#); unverified UI details flagged, not guessed.

v1.0

VCF Operations 9.0 / 9.1

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INTERNAL

Runbook — Implementing Capacity Reporting & Forecast Views in VCF Operations 9

Prepared by: Virtual Control LLC **Date:** July 31, 2026 **Platform:** VMware Cloud Foundation **Operations** 9.0 / 9.1 (formerly Aria Operations)

Legend: [V] = navigation/behaviour confirmed against a cited Broadcom TechDoc · [V-LIVE] = confirmed pattern, but verify the exact label in *your* build · [UNVERIFIED] = Broadcom page not reachable/didn't confirm — do not state as fact; verify in the UI first.

Goal: stand up **repeatable capacity reporting** (Time Remaining, Capacity Remaining, Reclaimable) and **forecast views** (trend + projection) that land in a **scheduled report** and a **dashboard** operators actually use — not just the out-of-box screens.

1. What "good" looks like (the deliverable)

- A **tuned capacity policy** (model, buffer, Time-Remaining thresholds) so the numbers reflect *your* risk posture.
- **Forecast Trend views** projecting each key resource forward, and **Time-Remaining / Capacity-Remaining list views** ranking the most-constrained objects.
- Reporting **super metrics** (e.g., per-volume percent-free, efficiency ratio) surfaced as decision-ready signals.
- A **dashboard** for at-a-glance capacity + a **scheduled PDF/CSV report** emailed to stakeholders.
- **What-If plans** for "can we land this growth / lose this host" questions.

2. Prerequisites

- **VCF Operations 9.x** healthy; the vCenter/vSAN/NSX (and any storage) **adapters collecting** — capacity/forecast are only as good as the data.
- **History for forecasting:** Trend views forecast from historical data — **default 90 days** [V]. New objects won't forecast meaningfully until enough history accrues. (*Trend View doc.*)
- **SMTP configured** in VCF Operations if you will **email** scheduled reports [V] (*email requires SMTP; exact recipient/schedule dialog is [UNVERIFIED] — see §7.*)
- Rights to edit **Policies, Views, Reports, Dashboards, Super Metrics** (Infrastructure Operations).

3. Phase 1 — Tune the capacity policy (do this FIRST)

Why first: the policy defines how Capacity Remaining and Time Remaining are *calculated* and when they go critical — reporting on untuned defaults gives numbers you can't stand behind.

3.1 Open the policy (verified-live navigation)

Configurations → **Policy Definition** → in the policy list select the policy (e.g., **Default Policy**, child of **Base Settings**) → **EDIT POLICY** (top-right) → **Capacity** section. **[V-LIVE]**

[V-LIVE] Inheritance / the padlock. A policy inherits from its parent (e.g., Default Policy ← Base Settings). Where a setting is inherited it shows a **lock icon** , and the section reads "**Local: 0 Policy elements / No Local Changes.**" That means **inherited, not disabled**. To change a value you **click the padlock to unlock (override locally)** — it then counts as a **Local** element on this policy.

3.2 The Capacity section has four sub-sections **[V-LIVE]**

Sub-section	Purpose	Status
Risk Level Configurations	How the engine projects Time Remaining — an Aggressive↔Conservative slider (levels) + a separate Peak focused checkbox	Fully observed — see 3.2.1 [V-LIVE]
Business Hours	The business-hours / peak window the engine considers	Collapsed — expand to configure; the one field set not yet captured live
Capacity Settings	Storage consideration · Allocation Model · Custom Profile · Capacity Buffer	Fully observed — see 3.2.2 [V-LIVE]
Capacity Thresholds and Metrics	Per-object-type criticality thresholds + custom metrics	Fully observed — see 3.3 [V-LIVE]

3.2.1 Risk Level Configurations → Set Risk Level (observed) **[V-LIVE]**

A **slider from Aggressive (left) ↔ Conservative (right)** with **discrete levels**, plus a separate **Peak focused** checkbox (a modifier combinable with any level). It is **not** three exclusive options.

Setting	How it projects Time Remaining	Use for
Conservative (levels increase → more conservative)	Based on the upper-bound projection of utilization (most headroom)	Production — performance critical
Aggressive	Based on the actual projected / mean utilization	Dev/Test — less performance-sensitive
Peak Focused (<i>checkbox</i>)	Additionally factors utilization spikes (regular/occasional) into planning	Spiky workloads; combinable with either level

- **Preview chart legend (observed):** Historical · Usable Capacity · Projection · Conservative (Upper Bound).
- **This lab's current value: Conservative Level 3**, Peak focused **unchecked**. **[V-LIVE]**
- **Recommendation for capacity reporting: Conservative** gives the safest, most defensible Time-Remaining numbers for production; enable **Peak focused** if the monitored workloads/storage spike.

3.2.2 Capacity Settings (observed) **[V-LIVE]**

Per object type (tree shows **Cluster Compute Resource · Datastore · Datastore Cluster**). Four lockable groups:

Group	Fields (observed)	Default
Storage consideration for capacity calculation	Exclude storage capacity toggle	Off (storage included)
Allocation Model	Enable via overcommit ratio per CPU / Memory / Disk Space (checkbox + ratio); Consider Powered off VMs → Activate	Off (Demand-only; Demand always calculates)
Custom Profile	Select properties dropdown ("how many more fit")	None selected
Capacity Buffer	Headroom % per Memory / CPU / Disk Space , split into Allocation and Demand columns	0% everywhere

- **Recommendation (production reporting):** set a **Demand Capacity Buffer** (commonly **10–20%** or per SLA) so Time/Capacity Remaining reserves headroom rather than reporting to 100%; leave **Allocation Model off** unless planning by overcommit; leave **Exclude storage** off; **Custom Profile is optional** — its "Select properties" dropdown can be **empty** (nothing to select) and it's not needed for capacity reporting, so skip it unless you want "how many more VMs fit."

[V-LIVE] Capacity Buffer behaviour (product tooltip): "Buffer reduces the usable capacity. In vSphere Cluster, it's applied after High Availability (HA). Use buffer as a safety net in capacity." → **usable = total – HA reservation – buffer**; the buffer stacks **on top of HA**. HA already reserves failover capacity, so **do NOT size the buffer to also cover a host loss** — it's pure extra margin (demand spikes / growth). A **10–15% Demand buffer** on top of HA is a defensible production default.

[V-LIVE] Disk-Space overcommit caveat: the Allocation Model's **Disk Space** ratio = your **thin-provisioning overcommit** (allocated vs physical). For **storage / production capacity reporting**, **do NOT overcommit disk** — set **1:1**, or (cleaner) leave the Allocation Model off and report on **Demand**. Overcommitting disk reports headroom that doesn't physically exist. A product ⓘ icon on the Disk Space line documents its specific behaviour — **read it before setting a ratio**. **[UNVERIFIED]:** exact ⓘ tooltip text (not captured).

Worked example — applied production config for this environment (Cluster Compute Resource): **[V-LIVE] - Capacity Buffer** → **Demand:** Memory **10%** · CPU **10%** · Disk Space **15%** · **Allocation columns 0** (*inert while Allocation Model is off — set the Demand column only*). - **Allocation Model:** off (Demand-only) · **Exclude storage capacity:** off · **Custom Profile:** empty (skipped). - **Risk Level:** Conservative **Level 3**, Peak focused **off**. - **Criticality Thresholds:** left at defaults — Time Remaining **90/120**, Capacity Remaining **5/10**, Workload **80/90/95** (*optionally raise Time-Remaining critical to 120 days for a full quarter of lead time*). - **Repeat** the Demand buffer on the **Datastore** object type for datastore-level headroom. **SAVE** to commit.

3.3 Capacity Thresholds and Metrics (observed in full) [V-LIVE]

- **Left pane — object-type tree.** Pick the type to configure. Under **vCenter:** Cluster Compute Resource · Custom Datacenter · Datacenter · Datastore · Datastore Cluster · Host System · Pod · Resource Pool · vCenter · Virtual Machine · vSphere Private World · vSphere World. Under **Container:** Department · Environment · Function · Licensing · Location · ... (*For capacity reporting, Cluster Compute Resource and Datastore are the key ones.*)
- **Two tabs: Criticality Thresholds** and **Custom Metrics**.

Criticality Thresholds — color-band sliders per object type. **Observed defaults for Cluster Compute Resource:**

Metric	Green (healthy)	Yellow (warning)	Red (critical)
Time Remaining	≥ 120 days	90–120	< 90 days (scale to 365)
Capacity Remaining	≥ 10%	5–10%	< 5% (scale to 100)
Workload	≤ 80%	80–90	≥ 90% (markers 80 / 90 / 95, scale to 100)

To set your own: **click the  on the metric to unlock**, then **drag the band markers** to your posture (e.g., raise Time-Remaining critical to 120 days for a production SLA).

Custom Metrics tab — **[V-LIVE]** per resource (**Memory / CPU / Disk Space**), it **replaces** the **Total Capacity** and **Utilization** metric the capacity engine uses (pencil  to swap). Defaults observed: Memory ...|Total Capacity (KB) / ...|Utilization (KB) ; CPU ...|Total Capacity (MHz) / ...|Demand (MHz) ; Disk Space ...|Total Capacity (GB) / ...|Utilization (GB) .

 **[V-LIVE] It REPLACES, it doesn't add.** Product banner: "Selecting any capacity metric will replace the default metrics. The replaced metric will be used for all the capacity calculations." One swap changes **all** capacity math for that resource/object type. **Leave the defaults for a standard vSphere cluster** — only override to base the engine on a different underlying metric (e.g., a storage metric on a storage object type).

Correcting an earlier note — where super metrics go: this tab substitutes the **Total-Capacity / Utilization inputs**, so it takes only **absolute** capacity/utilization metrics. A **percent-free / efficiency *ratio*** **does NOT belong here** (it's a %, not an absolute). Surface those super metrics in **Views and Reports** (§5/§7), where they're already computed. Use this tab only to drive the engine from a different absolute metric — e.g., NetApp storage metrics, set on the **NetApp object type** selected in the left tree.

3.4 Save & scope

- **SAVE** (bottom-left) commits; **CANCEL** discards. **[V-LIVE]**
- **Scope decision:** editing **Default Policy** applies to **everything inheriting it**. For a scoped posture, **ADD** a child policy (top-left of Policy Definition) and set Capacity there, then bind it to the target group — the existing "**Policy for Virtual Machines — Risk Profile 1/2/3**" entries are the model. **[V-LIVE]**

[V] Key gotcha: these capacity-policy settings drive the **Capacity engine** (Capacity page, per-object Capacity tab, Time/Capacity Remaining, dashboards). They do **NOT** affect the **trend/forecast math inside a View** — a Trend view forecasts the raw metric independently (§5). Know which forecast a stakeholder is looking at.

4. Phase 2 — Reporting super metrics (decision-ready signals)

Raw counters (bytes free, dedup counters) aren't reportable — normalize them.

Navigate: Infrastructure Operations → Configuring Super Metrics → create. [V]

- **Per-object Percent-Free** = $(\text{available} / \text{total}) \times 100$ — comparable across differently-sized objects; one threshold fleet-wide; trends/forecasts cleanly.

- **Efficiency Ratio** = `logical_used / physical_used` — shows whether storage efficiency is holding or degrading.
- Assign each super metric to its **object type via the policy** so it computes per-object, then it's selectable in Views/widgets like any metric.

[V] Super metrics can drive **custom capacity calculations** — Broadcom: "You can select the default metrics shipped with VMware Aria Operations, or create super metrics and select them for custom capacity calculations." (Aria Operations 8.18 Config Guide — the same capacity engine that ships in VCF Ops 9.x). They also behave as normal metrics in Views/widgets, and a **Trend view forecasts any selected metric**, so percent-free/efficiency-ratio trend and forecast cleanly — **surface those in Views & Reports (\$5/\$7)**, which is where ratio super metrics belong. To drive the capacity *engine* from a custom **absolute** metric, use **Policy** → **Capacity** → **Capacity Thresholds and Metrics** → **Custom Metrics** tab, which **replaces** the Total-Capacity/Utilization input per resource (see §3.3) — **note**: a *ratio* like percent-free is NOT valid there. **[UNVERIFIED]**: using a super metric as the engine's *Demand input* specifically.

5. Phase 3 — Build the views (the heart of the report)

Navigate: Infrastructure Operations → Views → Create. [V] Six view types: **List, Summary, Trend, Distribution, Text, Image.** [V]

Know which forecast you're showing (two different engines): - **The capacity engine** (Time Remaining / Capacity Remaining on the Capacity page, per-object Capacity tab, dashboards) forecasts with **real-time predictive analytics** — Broadcom names the methods: **linear regression + ARMA** (linear), **FFTs / pulses / wavelets** (periodic), plus **change-point detection, cyclical analysis**, and an **exponential-decay window**. Inputs = **Demand + Usable Capacity; projection window = 1 year.** [V] (Aria 8.18 Config Guide — same engine as VCF Ops 9.x.) - **A View's Trend forecast** is a **separate mechanism** — computed **solely from the resource–metric pair's historical data, default 90 days**; its specific algorithm is **not named** by Broadcom ([UNVERIFIED] — do **not** attribute regression/ARMA to a View forecast).

5.1 Forecast Trend view (the projection)

1. **Create** → **Trend** view. [V]
2. **Subjects**: the object type (e.g., Cluster Compute Resource, Datastore, or your storage object).
3. **Data**: pick the metric(s) to project — CPU/Mem/Disk usage, **Capacity Remaining**, or your **percent-free super metric**.
4. **Configuration**: enable **historical data + trend of historical data + forecast for future time**; set a **future end date** — the with-no-data future window is **populated by the forecast**. Add **threshold lines** (e.g., 90% or your buffer line). [V] (Default history window 90 days.)
5. **Create**.

5.2 Time-Remaining / Capacity-Remaining list view (the ranking)

1. **Create** → **List** view. [V]
2. **Subjects**: Cluster (and/or Datastore).
3. **Data**: add **Time Remaining, Capacity Remaining** (CPU/Mem/Disk), and **percent-free**; sort ascending on Time Remaining so the **most-constrained objects rank to the top.** [V-LIVE] (List views surface these metrics;

confirm exact metric-key labels in your build.)

4. Create.

5.3 (Optional) Reclaimable view

- A **List** view on **Reclaimable Capacity / Reclaimable VMs** to report recoverable resource + savings alongside the forecast. **[V]** (*Reclaimable is a first-class capacity concept; also visible on Capacity → Assess → VIEW RECLAIMABLE VMS.*)

6. Phase 4 — Dashboard (at-a-glance)

Two paths — use both:

A. Reuse the predefined Capacity Dashboards — start here, don't rebuild what ships. **[V]** The 9.x predefined capacity dashboards are:

Cluster Capacity · Datastore Capacity · ESX(i) Capacity · VM Capacity · Reclamation · vSAN Capacity · vSAN Stretched Clusters · vSAN ESA Capacity · vSAN OSA Capacity

B. Build a custom dashboard and place your \$5 views on it. Use these **verified 9.x widgets** (exact names):

Widget	Use
View	Place an existing \$5 view (its Output Data lists existing views — drag the View widget; don't use the view-type picker, which <i>creates</i> a new view)
Time Remaining	Days-to-full per object
Capacity Remaining	Remaining CPU/mem/disk
Object List / Top-N	Rank the most-constrained objects
Scoreboard	Single-number KPIs (e.g., worst Time Remaining)
Metric Chart	Live trend of a chosen metric / super metric
Efficiency	Reclamation / efficiency context

[V] Naming caution: there is **no** widget literally named "Capacity Utilization," "Forecast," or "Reclaimable Capacity" in 9.x — use **Time Remaining / Capacity Remaining / Efficiency** and the **Reclamation** dashboard instead. (Widget list confirmed against the Widget Definitions List.)

7. Phase 5 — Scheduled report (the deliverable that lands in an inbox)

Navigate: Infrastructure Operations → Dashboards & Reports → Reports → Create. **[V]**

Build the template: 1. **Name and Description.** [V] 2. **Report Content** — drag your **Views and Dashboards** (\$5–\$6) from the right pane into the **Report Template Structure**; per item set portrait/landscape and list colorization via the **:** menu. [V] 3. **Layout and Format** → choose **PDF and/or CSV** here (format is set on the *template*, not the schedule) → **Create.** [V]

Schedule + email it (exact dialog): [V] 1. **Dashboards & Reports** → **Reports** → **Report Template** tab → select your template → **:** (**vertical ellipsis**) → **Schedule** → select an object → **Next.** 2. **Time settings:** Time zone · Date · Hour · **Minutes (0 / 15 / 30 / 45).** 3. **Recurrence: Daily / Weekly / Monthly**, with adjustable periodicity (e.g., every 2 weeks). 4. **Email:** check **Email report** → **Email addresses** box → **CC list / BCC list** → select an **outbound rule.** 5. **Prerequisite:** email requires **Outbound Alert Settings** configured — the **Standard Email plug-in** under **Infrastructure Operations** → **Configuring Alerts and Actions** → **Outbound Settings.** [V] (*Only users created in VCF Operations can add/edit schedules; large reports generate sequentially and can take a while.*)

[UNVERIFIED] the SMTP **field-level** detail (host/port/user) lives on the "Adding Outbound Notification Plug-Ins" sub-page — confirm live. And Broadcom's 9.x docs **do not enumerate predefined *capacity report template* names** — do **not** state specific template names as fact; read the actual list in your instance (Reports → Report Template tab).

8. Phase 6 — Forecast planning with What-If Analysis

For "can we absorb this growth?" / "can we lose a host?" questions:

Navigate: **Capacity** → **NEW** → opens the **What-if Analysis Plan** page (create / save / compare scenarios). [V]

Scenario types [V]: Workload Planning (**Traditional; Hyperconverged / VMC on AWS**) · Infrastructure Planning (**Traditional** add/remove hosts; **Hyperconverged** add/remove vSAN nodes) · Migration Planning (**VMware Cloud; Public Cloud**) · **Data Center Comparison.**

(9.1: *What-If scales to 10,000 VMs / 200 TB, with updated forecast algorithms for accuracy/explainability.* [V])

9. The two built-in capacity views (both exist in 9.x) [V]

- **Centralized** — **Capacity page:** left menu → **Capacity** → **Assess** (datacenters/clusters), with **CAPACITY CONFIGURATION DETAILS, RECLAIM RESOURCES,** and **VIEW RECLAIMABLE VMS.** Start here to assess and reclaim.
- **Per-object** — **Capacity tab:** left menu → **Inventory** → **select the object** → **Details** → **Capacity** tab. It shows **Time Remaining, Capacity Remaining, Virtual Machine Remaining,** and per-resource **CPU / Memory / Disk Space** (Total, Usable, Used, Recommended Size, Remaining, Buffer, HA). Use this to drill into a single cluster/host/datastore.

[V] Both paths are confirmed — capacity is **not** only centralized. The per-object **Capacity tab** is the fast drill-down; **Capacity** → **Assess** is the fleet-wide assess/reclaim/what-if launch point.

10. Verify

- **Policy:** target group's Capacity Details reflect your model/buffer/Time-Remaining thresholds; Assess page numbers move accordingly.

- **Super metrics:** compute per-object (non-null) and are selectable in a View.
- **Trend view:** shows history **and** a forward forecast line to your future end date, with threshold line.
- **List view:** ranks objects by Time Remaining; percent-free present.
- **Dashboard:** views render via the View Widget.
- **Report:** generates **PDF + CSV**; a scheduled run produces output (and emails if SMTP set).
- **What-If:** a saved scenario returns a projection.

11. Sources (Broadcom TechDocs, VCF Operations 9.0/9.1)

- Cost & Capacity Management (overview) — [.../9-0/cost-and-capacity-management.html](#)
- How Does Capacity Optimization Work; Allocation & Demand Model — [.../capacity-optimization-concepts/...](#)
- Using the Capacity Page (Assess / Reclaim / Reclaimable VMs) — [.../how-to-optimize-capacity.../using-the-capacity-page-to-asses-and-optimize-capacity.html](#)
- What-If Analysis (Capacity → NEW; scenario types) — [.../how-to-plan-for-capacity-changes/what-if-analysis-modeling-workload-capacity-or-migration.html](#)
- Configuring Policies → Capacity Details (Time Remaining / Allocation Model / Capacity Buffer elements) — [.../infrastructure-operations/configuring-policies/.../policy-workspace-capacity-details/...](#)
- Views (Create; List; **Trend** — forecast lines, 90-day history, "capacity settings don't affect view forecast") — [.../reports-and-views/using-vcops-views/create-and-modify-a-view/...](#)
- Reports (Create template; PDF/CSV) — [.../reports-and-views/using-vcops-reports/...](#)
- Capacity Dashboards; View Widget; Metric Chart Widget — [.../infrastructure-operations/.../widget-definitions-list/...](#)
- Configuring Super Metrics — [.../infrastructure-operations/configuring-super-metrics.html](#)
- **Predefined Capacity Dashboards** (the 9 names) — [.../infrastructure-operations/predefined-dashboards-in-vrealize-operations-manager/capacity-dashboards.html](#)
- **Widget Definitions List** (exact widget names) — [.../infrastructure-operations/dashboards-and-widgets/using-widgets/widget-definitions-list.html](#)
- **Schedule a Report** (schedule + email dialog) — [.../reports-and-views/using-vcops-reports/manage-report-templates/schedule-a-report.html](#)
- **Outbound Settings** (Standard Email plug-in prereq) — [.../infrastructure-operations/configuring-alerts-and-actions/outbound-settings.html](#)
- **How to View and Assess Capacity** (per-object **Capacity tab** + Capacity page) — [.../cost-and-capacity-management/optimizing-capacity-and-improving-performance/how-to-view-and-assess-capacity.html](#)
- **Aria Operations 8.18 Config Guide** — "**How ... Calculate and Forecast Capacity**" (engine **algorithms**: linear regression/ARMA, FFT/wavelets, change-point, exponential decay; **1-yr** window; **super metrics for custom capacity calculations**) — [techdocs.broadcom.com/.../aria-operations/8-18/.../how-does-vmware-aria-operations-calculate-and-forecast-capacity.html](#)
- 9.1 What's New — VCF Ops (updated capacity algorithms; What-If 10k VMs/200 TB) — [.../9-1/release-notes/.../whats-new-vcf-ops.html](#)

Still UNVERIFIED — the only items to confirm live (everything else above is Broadcom-cited): 1.

Predefined *capacity report template* names — Broadcom's 9.x docs confirm templates exist but do **not** list them by name; read the list in your instance (Reports → Report Template tab). Do not state names as fact. 2.

The View Trend-forecast algorithm — unnamed by Broadcom. Don't attribute regression/ARMA to a *View* forecast (those names belong to the *capacity engine*). 3. **SMTP field-level detail** (host/port/user) — on the "Adding Outbound Notification Plug-Ins" sub-page; confirm the exact fields. 4. **Super metric as the engine's Demand input specifically** — the doc says "custom capacity calculations" generally; the Demand-input case isn't spelled out. 5. The **algorithm names come from the Aria 8.18 Config Guide** (same engine as VCF Ops 9.x) — cite it as such, since the VCF-9.x-branded page is high-level only.

Document Information

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Revision History

Version	Date	Notes
1.0	2026-07-31	Initial runbook: policy tuning → reporting super metrics → forecast Trend + Time-Remaining List views → dashboard (predefined + View Widget) → scheduled PDF/CSV report → What-If planning. All navigation Broadcom-cited; UI details that couldn't be confirmed flagged UNVERIFIED (dashboard/widget names, report schedule dialog, per-object capacity tab).

Document End